

PAVAN DHARMA ADAPA

Healthcare Data Analyst

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PROFESSIONAL SUMMARY

Healthcare data analyst with an M.S. in Computer Science. Builds the reporting, data management and validation that clinical and revenue-cycle decisions rest on. Analysis of real CMS claims filings and CDC survey microdata at national scale, with documented methods and stated limitations. Work-authorized through June 2027 on F-1 OPT and eligible for the 24-month STEM extension: no sponsorship required until 2029. Available immediately.

TECHNICAL SKILLS

Analysis & Reporting: SQL, Python (pandas, NumPy, scikit-learn), R, Advanced Excel, Power BI, Tableau

Healthcare Data: Claims and encounter data, revenue cycle (denials and appeals), quality measures (HEDIS, CMS eCQM), SNOMED CT, LOINC, ICD-10, HIPAA

Data Management: Data procurement and validation, quality control, dimensional modelling, reproducible pipelines, technical documentation, Git

PROFESSIONAL EXPERIENCE

Data Analytics Intern | *Agricultural Microbiomes Lab, Mississippi State University, Starkville, MS* Jul 2026 – Present

- Rebuilt the lab's DNA sequencing pipeline into a documented, tested tool that installs in one command — **cutting setup from several days to under an hour** — across 30 merged pull requests covering continuous integration, packaging, and cross-platform support.
- Validated pipeline output against an independently built implementation, **agreeing to within 0.01 percentage points** on every species and genus — the verification that made the lab's results defensible to reviewers.

Graduate Research Assistant | *Mississippi State University, Starkville, MS* Aug 2024 – Aug 2025

- Managed and analysed a **900,000+ record** public-health surveillance dataset, and prepared the written and oral reports presented to faculty across two departments.
- Built the automated collection and cleaning workflow that standardised multi-source research data, **cutting per-dataset processing from hours to under 10 minutes**.

Graduate Teaching Assistant | *Mississippi State University, Starkville, MS* Sep 2025 – May 2026

- Taught data cleaning, transformation and visualisation in Python and R to **100+ students across six courses**, and reported completion and grade metrics that let instructors reach at-risk students earlier.

Marketing Data Analyst Intern | *Viral Fission, Hyderabad, India* May 2023 – Dec 2023

- Built automated Power BI and GA4 reporting for campaign performance, **halving manual reporting time** for a team running national brand campaigns.

HEALTHCARE ANALYTICS PROJECTS

Claims Denials & Appeals — *CMS Transparency in Coverage, 2021–2023*

- Built a **15,545 plan-year** SQL panel covering **126.9M denied claims** across every federal Marketplace plan, using window functions over a two-grain fact design.
- Found only **0.168% of denials are ever appealed while 40.1% of appeals are overturned** — quantifying the recoverable revenue a revenue-cycle team leaves unclaimed.

Clinical Quality Measures Engine — *HEDIS / CMS eCQM · synthetic non-PHI EHR (Synthea)*

- Computed HEDIS and CMS eCQM measures over a **13,810-patient EHR** (820,993 encounters, 10.8M observations), holding each measure as a versioned specification with its SNOMED CT and LOINC value sets separate from the engine — so an annual steward update is a reviewable diff a clinical analyst signs off, not a code change.
- Decomposed a diabetes control measure to show **203 of 204 failures were patients never tested**, reframing it as a testing-outreach problem rather than a glycemic-control one.

EDUCATION

M.S., Computer Science — Mississippi State University, Mississippi State, MS May 2026

B.Tech, Computer Science — Malla Reddy University, Hyderabad, India May 2024

Publication (in progress): Zhang Li, Pavan Dharma Adapa, Sai Harika Gade, S. Ratnavale, N. Michael — Computational Tools for Modeling Respiratory Disease Spread in Commercial Poultry Production.